Grant overview

Identification

- **Programme:** ITRI ICL 115-year subcontract / collaborative research, public notice no. 5.
- **Title:** Semantic-Spatial Translation and Safety-Constrained VLA for ArduPilot UAVs.
- **Period:** 2026-02-01 – 2026-11-30 (Republic of China year 115).
- **Budget:** NT$ 1,012,000.
- **PI:** Kuan-Ting Lai, NTUT AIoT Lab.
- **Personnel:** 1 PI + 4 Master students (two on ArduPilot/MAVLink, two on VLA fine-tuning / prompt engineering).

Work packages

ITRI defines the top-level system spec; NTUT executes WP1 through WP4.

WP	Title	Output artefact	Owns these KPIs
WP1	Policy / constraint IR / DSL	YAML/JSON DSL → IR/AST; signed policy bundle (hash + semver) covering GeoFence, envelope, corridor, time windows, breach actions	Policy load round-trip; bundle replayability
WP2	Prefix Constraint Compiler	Constraint Summary Pack (CSP) injected into VLA / planner prefix	Prefix-token budget; CSP coverage of policy IR
WP3	Suffix Safety Shield	ROS 2 node: Monitor / Filter / Projection; MAVLink → ArduPilot fence / RTL / Land / Loiter as ultimate backstop	**P0 violation escape rate (target 0)**; **fail-safe trigger correctness**; **mean repair magnitude**; **mean time to safe**

WP	Title	Output artefact	Owns these KPIs
WP4	Dynamic-scenario stress testing	Scenario library; KPI rollups; replay harness	Mission success rate; per-paraphrase robustness

The sub-step-level implementation breakdown for each component lives in the Implementation plan section.

Acceptance KPIs (explicitly listed)

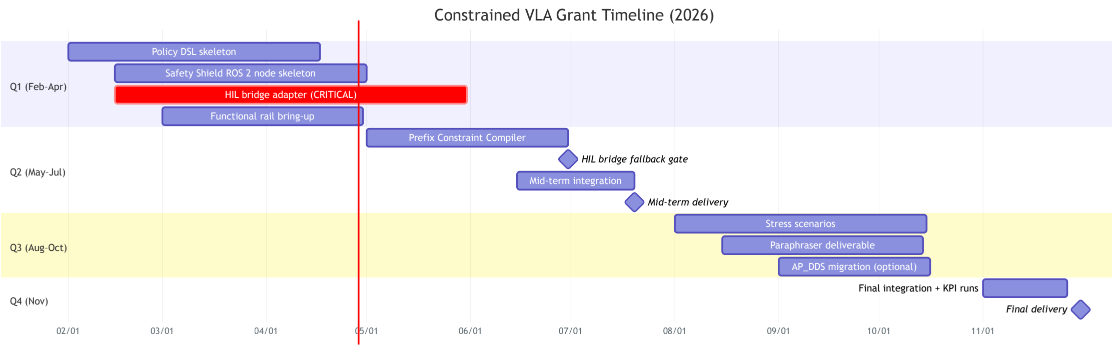
1. Mission success rate (任務成功率).
2. **P0 violation escape rate — target 0.**
3. Fail-safe trigger correctness.
4. Mean repair magnitude.
5. Mean time to safe.

Delivery dates

The two dates below are **hard contractual acceptance gates**.

Date	Item	What ships
2026-07-20	Mid-term delivery	Policy DSL + IR; Safety Shield ROS 2 node skeleton with at least one projection operator; functional rail (Gazebo Harmonic) demoable end-to-end
2026-11-30	Final delivery	Policy DSL, Prefix Compiler, Safety Shield, and Stress Testing complete; KPI report on dynamic-scenario stress; perception-rail integration; signed final report

Timeline



Locked architectural constraints

These are not re-litigated in this report; they are inputs to the simulation choice.

- **Hierarchical control:** the VLA outputs a setpoint or mission item; ArduPilot runs the inner loop at 100–400 Hz. End-to-end motor-PWM VLAs are out of scope for this grant.
- **Action space:** 4-D `(vx, vy, vz, yaw_rate)`. This is the Safety Shield's input contract — projection operators in the Safety Shield act on this 4-D space, not on waypoint-shaped actions. The shape was chosen for compatibility with current drone-VLA work (CognitiveDrone is one such baseline reference); the project does not commit to a specific backend as the "default".
- **Safety path:** MAVROS 2 first; AP_DDS migration is a Q3 optional, not a blocker.
- **GCS:** Mission Planner via `mavlink-router` fan-out (parallel to MAVROS), not as a serial bottleneck.

See Architecture constraints for the diagram.